** PROJECT DAY - 28 **

[Spam-Mail Prediction Using ML]

Importing the Dependencies

In [204…
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

Data Collection & Pre-Processing

In [206…
```
# loading the data from csv file to a pandas Dataframe
raw_mail_data = pd.read_csv('mail_data.csv')
```

In [207…
```
print(raw_mail_data)
```

```
    Category                                            Message
0        ham  Go until jurong point, crazy.. Available only ...
1        ham                      Ok lar... Joking wif u oni...
2       spam  Free entry in 2 a wkly comp to win FA Cup fina...
3        ham  U dun say so early hor... U c already then say...
4        ham  Nah I don't think he goes to usf, he lives aro...
...      ...                                                ...
5567    spam  This is the 2nd time we have tried 2 contact u...
5568     ham               Will ü b going to esplanade fr home?
5569     ham  Pity, * was in mood for that. So...any other s...
5570     ham  The guy did some bitching but I acted like i'd...
5571     ham                         Rofl. Its true to its name

[5572 rows x 2 columns]
```

In [208…
```
# replace the null values with a null string
mail_data = raw_mail_data.where((pd.notnull(raw_mail_data)),'')
```

In [209…
```
# printing the first 5 rows of the dataframe
mail_data.head()
```

Out[209…

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

In [210…
```
# checking the number of rows and columns in the dataframe
mail_data.shape
```

Out[210…
```
(5572, 2)
```

Label Encoding

In [212…
```
# label spam mail as 0;  ham mail as 1;

mail_data.loc[mail_data['Category'] == 'spam', 'Category',] = 0
mail_data.loc[mail_data['Category'] == 'ham', 'Category',] = 1
```

spam - 0
ham - 1

In [214…
```
# separating the data as texts and label

X = mail_data['Message']

Y = mail_data['Category']
```

In [215…
```
X
```

Out[215…
```
0       Go until jurong point, crazy.. Available only ...
1                           Ok lar... Joking wif u oni...
2       Free entry in 2 a wkly comp to win FA Cup fina...
3       U dun say so early hor... U c already then say...
4       Nah I don't think he goes to usf, he lives aro...
                              ...
5567    This is the 2nd time we have tried 2 contact u...
5568                 Will ü b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571                           Rofl. Its true to its name
Name: Message, Length: 5572, dtype: object
```

In [225…
```
Y
```

Out[225…
```
0       1
1       1
2       0
3       1
4       1
       ..
5567    0
5568    1
5569    1
5570    1
5571    1
Name: Category, Length: 5572, dtype: object
```

Splitting the data into training data & test data

In [227…
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=3)
```

In [228…
```
print(X.shape)
print(X_train.shape)
print(X_test.shape)
```

```
(5572,)
(4457,)
(1115,)
```

Feature Extraction

In [230…
```
from sklearn.feature_extraction.text import TfidfVectorizer

X_train = X_train.fillna("").astype(str)
X_test = X_test.fillna("").astype(str)

vectorizer = TfidfVectorizer(
    min_df=1,
    stop_words='english',
    lowercase=True
)

X_train_features = vectorizer.fit_transform(X_train)
X_test_features = vectorizer.transform(X_test)
```

In [235…
```
print(X_train)
```

```
3075                  Don know. I did't msg him recently.
1787    Do you know why god created gap between your f...
1614                           Thnx dude. u guys out 2nite?
4304                                        Yup i'm free...
3266    44 7732584351, Do you want a New Nokia 3510i c...
                              ...
789     5 Free Top Polyphonic Tones call 087018728737,...
968     What do u want when i come back?.a beautiful n...
1667    Guess who spent all last night phasing in and ...
3321    Eh sorry leh... I din c ur msg. Not sad alread...
1688    Free Top ringtone -sub to weekly ringtone-get ...
Name: Message, Length: 4457, dtype: object
```

In [237…
```
print(X_train_features)
```

```
  (0, 2329)	0.38783870336935383
  (0, 3811)	0.34780165336891333
  (0, 2224)	0.413103377943378
  (0, 4456)	0.4168658090846482
  (0, 5413)	0.6198254967574347
  (1, 3811)	0.17419952275504033
  (1, 3046)	0.2503712792613518
  (1, 1991)	0.33036995955537024
  (1, 2956)	0.33036995955537024
  (1, 2758)	0.3226407885943799
  (1, 1839)	0.2784903590561455
  (1, 918)	0.22871581159877646
  (1, 2746)	0.3398297002864083
  (1, 2957)	0.3398297002864083
  (1, 3325)	0.31610586766078863
  (1, 3185)	0.29694482957694585
  (1, 4080)	0.18880584110891163
  (2, 6601)	0.6056811524587518
  (2, 2404)	0.45287711070606745
  (2, 3156)	0.4107239318312698
  (2, 407)	0.509272536051008
  (3, 7414)	0.8100020912469564
  (3, 2870)	0.5864269879324768
  (4, 2870)	0.41872147309323743
  (4, 487)	0.2899118421746198
  :	:
  (4454, 2855)	0.47210665083641806
  (4454, 2246)	0.47210665083641806
  (4455, 4456)	0.24920025316220423
  (4455, 3922)	0.31287563163368587
  (4455, 6916)	0.19636985317119715
  (4455, 4715)	0.30714144758811196
  (4455, 3872)	0.3108911491788658
  (4455, 7113)	0.30536590342067704
  (4455, 6091)	0.23103841516927642
  (4455, 6810)	0.29731757715898277
  (4455, 5646)	0.33545678464631296
  (4455, 2469)	0.35441545511837946
  (4455, 2247)	0.37052851863170466
  (4456, 2870)	0.31523196273113385
  (4456, 5778)	0.16243064490100795
  (4456, 334)	0.2220077711654938
  (4456, 6307)	0.2752760476857975
  (4456, 6249)	0.17573831794959716
  (4456, 7150)	0.3677554681447669
  (4456, 7154)	0.24083218452280053
  (4456, 6028)	0.21034888000987115
  (4456, 5569)	0.4619395404299172
  (4456, 6311)	0.30133182431707617
  (4456, 647)	0.30133182431707617
  (4456, 141)	0.292943737785358
```

Training the Model

Logistic Regression

In [240…
```
model = LogisticRegression()
```

In [247…
```
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
import pandas as pd

# ensure pandas Series
Y_train = pd.Series(Y_train)
Y_test = pd.Series(Y_test)

# handle missing values (future-proof)
Y_train = Y_train.fillna(0).infer_objects(copy=False).astype(int)
Y_test  = Y_test.fillna(0).infer_objects(copy=False).astype(int)

# encode labels if text (safety check)
if Y_train.dtype == 'object':
    encoder = LabelEncoder()
    Y_train = encoder.fit_transform(Y_train)
    Y_test = encoder.transform(Y_test)

# train model
model = LogisticRegression(max_iter=1000)
model.fit(X_train_features, Y_train)
```

Out[247…
```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

Evaluating the trained model

In [250…
```
# prediction on training data

prediction_on_training_data = model.predict(X_train_features)
accuracy_on_training_data = accuracy_score(Y_train, prediction_on_training_data)
```

In [252…
```
print('Accuracy on training data : ', accuracy_on_training_data)
```

```
Accuracy on training data :  0.9676912721561588
```

In [254…
```
# prediction on test data

prediction_on_test_data = model.predict(X_test_features)
accuracy_on_test_data = accuracy_score(Y_test, prediction_on_test_data)
```

In [256…
```
print('Accuracy on test data : ', accuracy_on_test_data)
```

```
Accuracy on test data :  0.9668161434977578
```

Building a Predictive System

In [259…
```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

# Vectorizer (fit here)
feature_extraction = TfidfVectorizer(
    min_df=1,
    stop_words='english',
    lowercase=True
)

X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)

# Model
model = LogisticRegression(max_iter=1000)
model.fit(X_train_features, Y_train)
```

Out[259…
```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

In [261…
```
input_mail = [
    "I've been searching for the right words to thank you for this breather..."
]

input_data_features = feature_extraction.transform(input_mail)
prediction = model.predict(input_data_features)

print(prediction)

if prediction[0] == 1:
    print("Ham mail")
else:
    print("Spam mail")
```

```
[1]
Ham mail
```

** END - PROJECT **

In [ ]: